Lab 22: Give synthesis one narrow tool, not the whole web

Topic 2: Tool Design & MCP Integration · Task 2.3: Distribute tools appropriately across agents and configure tool choice

Runs on Simulator 3: Multi-Agent Research System · Self-practice

Source: Official exam guide, sample question 9

Commit to an answer first

During testing, you observe that the synthesis agent frequently needs to verify specific claims while combining findings. Currently, when verification is needed, the synthesis agent returns control to the coordinator, which invokes the web search agent, then re-invokes synthesis with results. This adds 2-3 round trips per task and increases latency by 40%. Your evaluation shows that 85% of these verifications are simple fact-checks (dates, names, statistics) while 15% require deeper investigation. What's the most effective approach to reduce overhead while maintaining system reliability?

A. Give the synthesis agent a scoped verify_fact tool for simple lookups, while complex verifications continue delegating to the web search agent through the coordinator.

B. Have the synthesis agent accumulate all verification needs and return them as a batch to the coordinator at the end of its pass, which then sends them all to the web search agent at once.

C. Give the synthesis agent access to all web search tools so it can handle any verification need directly without round-trips through the coordinator.

D. Have the web search agent proactively cache extra context around each source during initial research, anticipating what the synthesis agent might need to verify.

Simulate the solution

```
# In the browser: open Simulator 3, then Lab 22
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 22

# In a terminal, from simulators/simulator-3-multi-agent-research/
uv run research_system.py
```

What you should see

Synthesis calls verify_fact. No agent uses a tool outside its definition.

Open `simulators/simulator-3-multi-agent-research/evidence/` and find it yourself.

The answer

A. A scoped verify_fact tool for simple lookups. Complex cases still go through the hub.

Option A applies the principle of least privilege by giving the synthesis agent only what it needs for the 85% common case (simple fact verification) while preserving the existing coordination pattern for complex cases. Option B's batching approach creates blocking dependencies since synthesis steps may depend on earlier verified facts. Option C over-provisions the synthesis agent, violating separation of concerns. Option D relies on speculative caching that cannot reliably predict what the synthesis agent will need to verify.

Now change one thing

In fixtures/sources.json replace the visual-arts TIMEOUT with [] and rerun: a valid empty result must not read as a failure.